

The optimization problem for SVM

- The optimal hyperplane is a solution of the following constrained optimization problem.
- Find $W \in \mathbb{R}^m$, $b \in \mathbb{R}$ to

$$\text{minimize} \quad \frac{1}{2} W^T W$$

$$\text{subject to} \quad 1 - y_i(W^T X_i + b) \leq 0, \quad i = 1, \dots, n$$

- Quadratic cost function and linear (inequality) constraints.
- Kuhn-Tucker conditions are necessary and sufficient. Every local minimum is global minimum.



- The Lagrangian is given by

$$L(W, b, \boldsymbol{\mu}) = \frac{1}{2}W^T W + \sum_{i=1}^n \mu_i [1 - y_i(W^T X_i + b)]$$

- The Kuhn-Tucker conditions give

$$\nabla_W L = 0 \Rightarrow W^* = \sum_{i=1}^n \mu_i^* y_i X_i$$

$$\frac{\partial L}{\partial b} = 0 \Rightarrow \sum_{i=1}^n \mu_i^* y_i = 0$$

$$1 - y_i(X_i^T W^* + b^*) \leq 0, \quad \forall i$$

$$\mu_i^* \geq 0, \quad \& \quad \mu_i^* [1 - y_i(X_i^T W^* + b^*)] = 0, \quad \forall i$$



- Let $S = \{i \mid \mu_i^* > 0\}$.
- By complementary slackness condition,

$$i \in S \Rightarrow y_i(X_i^T W^* + b^*) = 1$$

Implies X_i is closest to separating hyperplane.

- $\{X_i \mid i \in S\}$ are called Support vectors. We have

$$W^* = \sum_i \mu_i^* y_i X_i = \sum_{i \in S} \mu_i^* y_i X_i$$

- Optimal W is a linear combination of Support vectors.
- Support vectors constitute a very useful output of the method.



The SVM solution

- The optimal hyperplane – W^* , b^* given by:

$$W^* = \sum_i \mu_i^* y_i X_i = \sum_{i \in S} \mu_i^* y_i X_i$$

$$b^* = y_j - X_j^T W^*, \quad j \text{ s.t. } \mu_j^* > 0$$

(Note that $\mu_j^* > 0 \Rightarrow y_j(X_j^T W^* + b^*) = 1$)

- Thus, W^*, b^* are determined by $\mu_i^*, i = 1, \dots, n$.
- We can use the dual of the optimization problem to get μ_i^* .



Dual optimization problem for SVM

- The dual function is

$$q(\boldsymbol{\mu}) = \inf_{W, b} \left\{ \frac{1}{2} W^T W + \sum_{i=1}^n \mu_i [1 - y_i (W^T X_i + b)] \right\}$$

- If $\sum \mu_i y_i \neq 0$ then $q(\boldsymbol{\mu}) = -\infty$.
- Hence we need to maximize q only over those $\boldsymbol{\mu}$ s.t. $\sum \mu_i y_i = 0$.
- Infimum w.r.t. W is attained at $W = \sum \mu_i y_i X_i$.
- We obtain the dual by substituting $W = \sum \mu_i y_i X_i$ and imposing $\sum \mu_i y_i = 0$.



- By substituting $W = \sum \mu_i y_i X_i$ and $\sum \mu_i y_i = 0$ we get

$$\begin{aligned}
 q(\boldsymbol{\mu}) &= \frac{1}{2} W^T W + \sum_{i=1}^n \mu_i - \sum_{i=1}^n \mu_i y_i (W^T X_i + b) \\
 &= \frac{1}{2} \left(\sum_i \mu_i y_i X_i \right)^T \sum_j \mu_j y_j X_j + \sum_i \mu_i \\
 &\quad - \sum_i \mu_i y_i X_i^T \left(\sum_j \mu_j y_j X_j \right) \\
 &= \sum_i \mu_i - \frac{1}{2} \sum_i \sum_j \mu_i y_i \mu_j y_j X_i^T X_j
 \end{aligned}$$

- Thus, the dual problem is:

$$\max_{\boldsymbol{\mu}} \quad q(\boldsymbol{\mu}) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j y_i y_j X_i^T X_j$$

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subject to $\mu_i \geq 0, \quad i = 1, \dots, n, \quad \sum_{i=1}^n y_i \mu_i = 0$

- Quadratic cost function and linear constraints
- Training data vectors appear only as innerproduct
- Optimization is over $\mathbb{R}^n$ irrespective of the dimension of X_i .



- The dual problem is:

$$\max_{\boldsymbol{\mu}} \quad q(\boldsymbol{\mu}) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j y_i y_j X_i^T X_j$$

subject to $\mu_i \geq 0, \quad i = 1, \dots, n, \quad \sum_{i=1}^n y_i \mu_i = 0$

- Then the final solution is:

$$W^* = \sum \mu_i^* y_i X_i, \quad b^* = y_j - X_j^T W^*, \quad j \text{ such that } \mu_j > 0$$



- So far, we assumed that the training data is linearly separable.
- What happens if the data is non-separable?
- Optimization problem has no feasible point (and hence no solution) if data are not linearly separable.
- Hence we can not find the optimal hyperplane by this formulation.
- We will modify the problem by introducing slack variables so that we can handle the general case.



Using slack variables

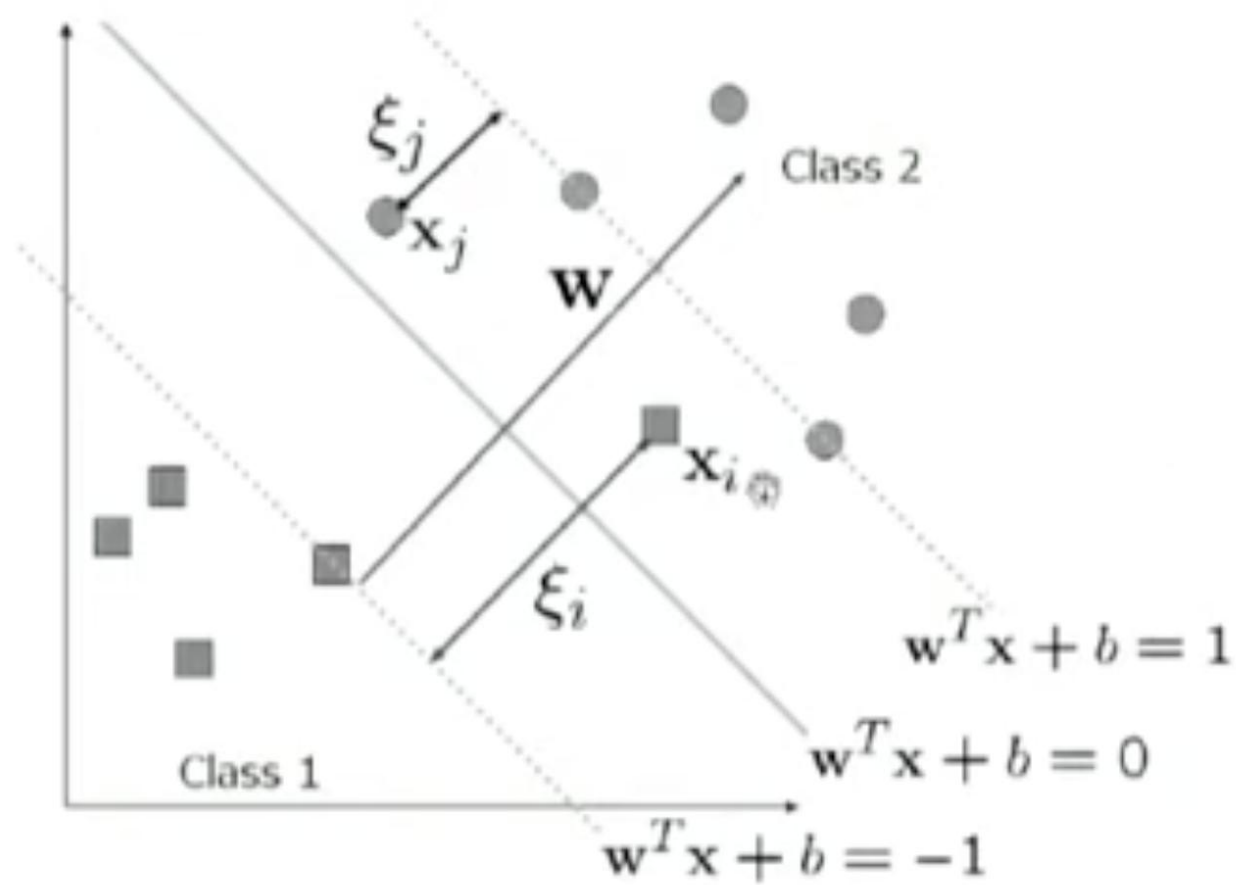
- When data are not linearly separable, we can try:

$$\text{minimize} \quad \frac{1}{2}W^TW + C \sum_{i=1}^n \xi_i$$

$$\text{subject to} \quad y_i(W^TX_i + b) \geq 1 - \xi_i, \quad i = 1, \dots, n$$
$$\xi_i \geq 0, \quad i = 1, \dots, n$$

- Opt. variables: W, b, ξ_i .





- The optimization problem now is

$$\min_{W, b, \xi} \quad \frac{1}{2} W^T W + C \sum_{i=1}^n \xi_i$$

$$\text{subject to} \quad \begin{aligned} 1 - \xi_i - y_i(W^T X_i + b) &\leq 0, \quad i = 1, \dots, n \\ -\xi_i &\leq 0, \quad i = 1, \dots, n \end{aligned}$$



- The Lagrangian now is

$$L(W, b, \xi, \mu, \lambda) = \frac{1}{2} W^T W + C \sum_{i=1}^n \xi_i \\ + \sum_{i=1}^n \mu_i (1 - \xi_i - y_i(W^T X_i + b)) - \sum_{i=1}^n \lambda_i \xi_i$$

- μ_i are the lagrange multipliers for the separability constraints as earlier.
- λ_i are the lagrange multipliers for the constraints $-\xi_i \leq 0$.



The Kuhn-Tucker conditions give us

- $\nabla_W L = 0 \Rightarrow W^* = \sum_{i=1}^{\ell} \mu_i^* y_i X_i$
- $\frac{\partial L}{\partial b} = 0 \Rightarrow \sum \mu_i^* y_i = 0$
- $\frac{\partial L}{\partial \xi_i} = 0 \Rightarrow \mu_i^* + \lambda_i^* = C, \forall i$
- $1 - \xi_i - y_i(W^T X_i + b) \leq 0; \quad \xi_i \geq 0; \quad \forall i$
- $\mu_i \geq 0; \quad \lambda_i \geq 0, \quad \forall i$
- $\mu_i(1 - \xi_i - y_i(W^T X_i + b)) = 0; \quad \lambda_i \xi_i = 0, \quad \forall i$



- The W^* is given by the same expression.
- We also have $0 \leq \mu_i + \lambda_i = C, \forall i$.
- If $0 < \mu_i < C$, then, $\lambda_i > 0$ which implies $\xi_i = 0$.
- Now the complementary slackness condition, we have $1 - y_i(W^T X_i + b) = 0$.
- Thus we get b^* as

$$b^* = y_j - X_j^T W^*, \quad j \text{ such that } 0 < \mu_j < C$$

- Thus, once again we need to find all μ_i .



- We can derive the dual as before. The dual function is

$$q(\mu, \lambda) = \inf_{W, b, \xi} L(W, b, \xi, \mu, \lambda)$$

where the lagrangian is given by

$$\begin{aligned} L(W, b, \xi, \mu, \lambda) = & \frac{1}{2} W^T W + C \sum_{i=1}^n \xi_i \\ & + \sum_{i=1}^n \mu_i (1 - \xi_i - y_i (W^T X_i + b)) - \sum_{i=1}^n \lambda_i \xi_i \end{aligned}$$



- In the lagrangian we have the term $\sum_i (C - \mu_i - \lambda_i)\xi_i$.
- Since we take infimum w.r.t. ξ_i , we need to impose $C = \mu_i + \lambda_i, \forall i$.
- When we impose this, all the terms containing λ_i or ξ_i drop out and hence now the q function would be same as earlier.
- We only need to ensure (in the dual) that $\lambda_i \geq 0$ and $C = \mu_i + \lambda_i, \forall i$.
- This is easily done by ensuring $0 \leq \mu_i \leq C$.



- The dual problem is:

$$\max_{\boldsymbol{\mu}} \quad q(\boldsymbol{\mu}) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j y_i y_j X_i^T X_j$$

subject to $0 \leq \mu_i \leq C, \quad i = 1, \dots, n, \quad \sum_{i=1}^n y_i \mu_i = 0$

- We solve dual and the final optimal hyperplane is

$$W^* = \sum \mu_i^* y_i X_i,$$

$$b^* = y_j - X_j^T W^*, \quad j \text{ such that } 0 < \mu_j < C.$$



- The primal problem is

$$\text{minimize} \quad \frac{1}{2}W^TW + C \sum_{i=1}^n \xi_i$$

$$\text{subject to} \quad y_i(W^TX_i + b) \geq 1 - \xi_i, \quad i = 1, \dots, n$$
$$\xi_i \geq 0, \quad i = 1, \dots, n$$

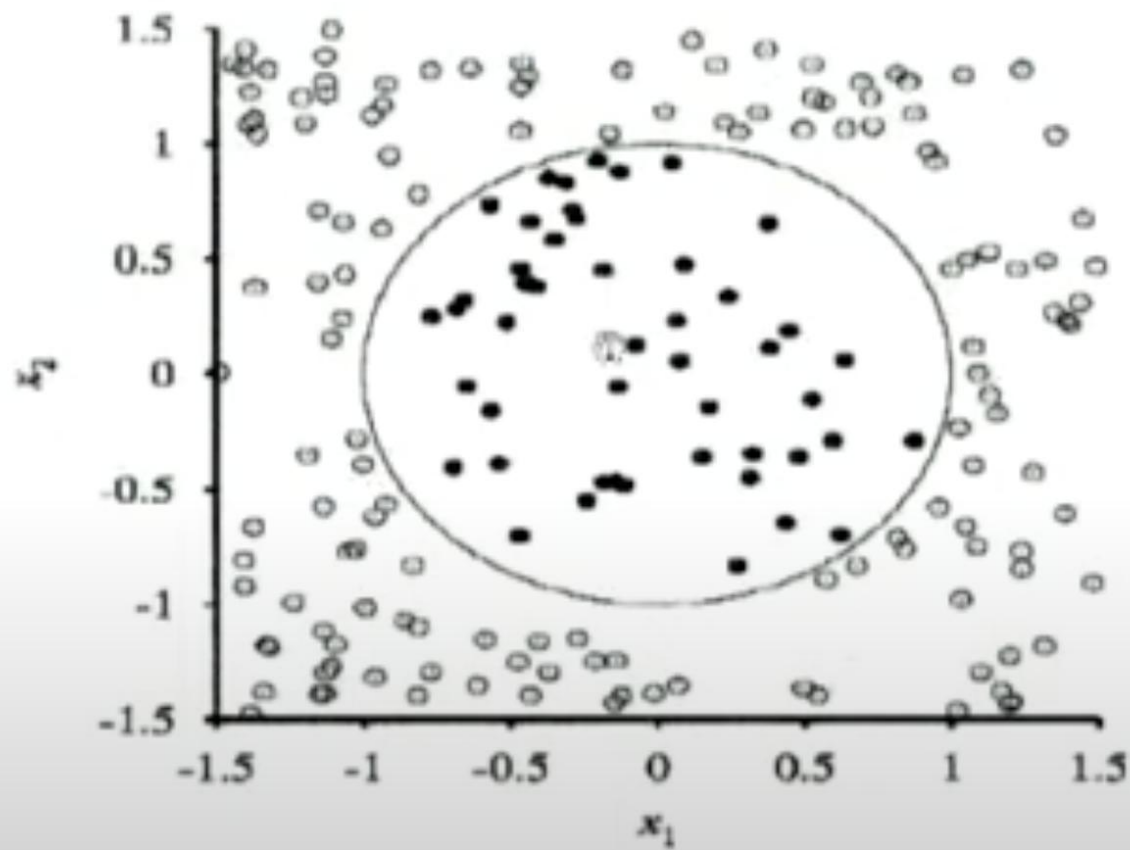
- As $C \rightarrow \infty$, we get back the old problem.



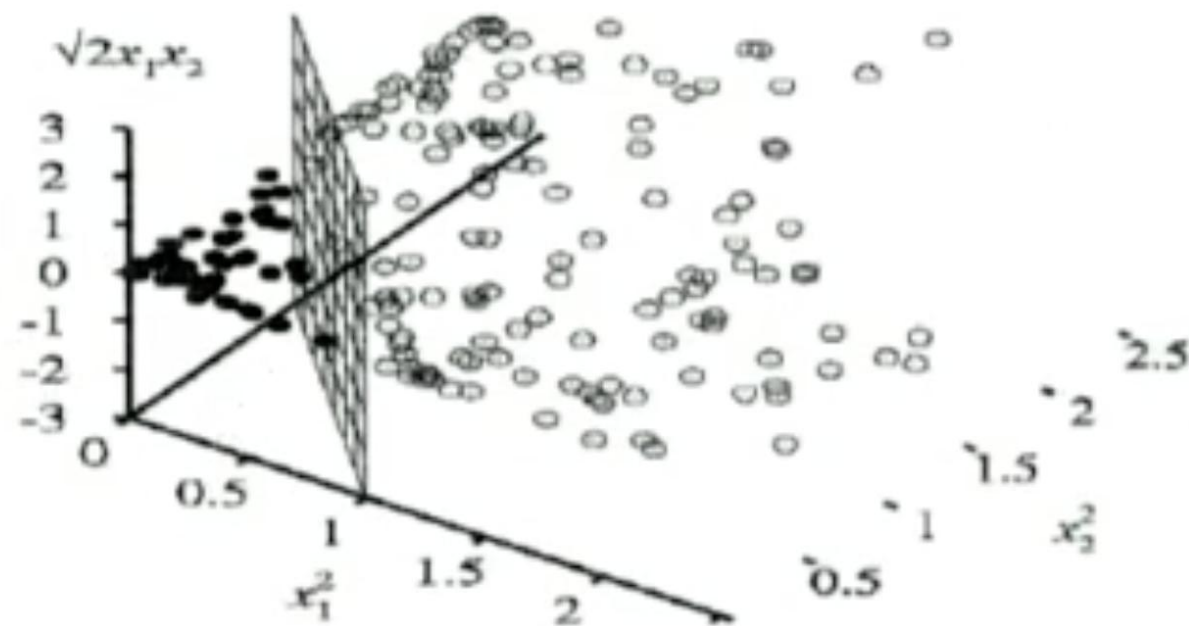
- By using slack variables, ξ_i , we can find 'best' hyperplane classifier.
- In the dual, the only difference is an upperbound on μ_i .
- The best linear classifier may not be 'good enough'
- How can we learn non-linear discriminant functions?
- Recall that the SVM idea is to transform X_i into some other high-dimensional space and learn a linear classifier there.



An example



An example



Non-linear discriminant functions

- In general, we can use a mapping, $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^{m'}$.
- In $\mathbb{R}^{m'}$, the training set is $\{(Z_i, y_i), i = 1, \dots, \ell\}$, $Z_i = \phi(X_i)$.
- We can find optimal hyperplane by solving the dual (replacing $X_i^t X_j$ with $Z_i^t Z_j$).
- The dual problem now would be the following.



$$\max_{\mu} \quad q(\mu) = \sum_{i=1}^n \mu_i - \frac{1}{2} \sum_{i,j=1}^n \mu_i \mu_j y_i y_j \phi(X_i)^T \phi(X_j)$$

subject to $0 \leq \mu_i \leq C, \quad i = 1, \dots, n, \quad \sum_{i=1}^n y_i \mu_i = 0$

- This is an optimization problem over $\mathbb{R}^n$ (with quadratic cost function & linear constraints) irrespective of ϕ and m' .
- But computationally expensive?



Kernel function

- Suppose we have a function, $K : \mathbb{R}^m \times \mathbb{R}^m \rightarrow \mathbb{R}$, such that

$$K(X_i, X_j) = \phi(X_i)^T \phi(X_j)$$

Called Kernel function.

- Suppose computation of $K(X_i, X_j)$ is about as expensive as that of $X_i^T X_j$.
- Replacing $Z_i^T Z_j$ by $K(X_i, X_j)$, we can solve dual without ever computing any $\phi(X_i)$. Efficient for obtaining optimal hyperplane.
- What about storing W^* ? Computing $\phi(X)^T W^*$ for new patterns?



Kernel function based classifier

- Let μ_i^* be soln of Dual. Then $W^* = \sum \mu_i^* y_i \phi(X_i)$.
- Then we have

$$b^* = y_j - \phi(X_j)^T W^* = (y_j - \sum_i \mu_i^* y_i \phi(X_i)^T \phi(X_j))$$

- Given a new pattern X we only need to compute

$$\begin{aligned} f(X) &= Z^T W^* + b^* \\ &= \sum_i \mu_i^* y_i \phi(X_i)^T \phi(X) + b^* \\ &= \sum_i \mu_i^* y_i K(X_i, X) + (y_j - \sum_i \mu_i^* y_i K(X_i, X_j)) \end{aligned}$$



- This is an interesting way of learning nonlinear classifiers.
- We solve the dual whose dimension is that of n , number of examples.
- All we need to store are:
 - non-zero Lagrange multipliers: $\mu_i^* > 0$,
 - Support vectors: X_i, i s.t. $\mu_i^* > 0$.
- Never need to enter ' $\phi(X)$ ' space!



Support Vector Machine

- Obtain μ_i^* by solving the Dual with $Z_i^T Z_j$ replaced by $K(X_i, X_j)$. (Choose a suitable Kernel function. Use 'penalty const', C if needed).
- Store non-zero μ_i^* and the corresponding support vectors.
- Classify any new pattern X by sign of
$$f(X) = \sum \mu_i^* y_i K(X_i, X) + (y_j - \sum_i \mu_i^* y_i K(X_i, X_j))$$
- If we have a suitable Kernel function, we never need to compute $\phi(X)$.
- The range space of ϕ can even be infinite dimensional!



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- Then we have (for j s.t. $0 < \mu_j < C$)

$$b^* = y_j - \phi(X_j)^T W^* = (y_j - \sum_i \mu_i^* y_i \phi(X_i)^T \phi(X_j))$$

- Given a new pattern X we only need to compute

$$\begin{aligned} f(X) &= Z^T W^* + b^* = \phi(X)^T W^* + b^* \\ &= \sum_i \mu_i^* y_i \phi(X_i)^T \phi(X) + b^* \\ &= \sum_i \mu_i^* y_i K(X_i, X) + (y_j - \sum_i \mu_i^* y_i K(X_i, X_j)) \end{aligned}$$



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$$f(X) = \sum \mu_i^* y_i K(X_i, X) + (y_j - \sum_i \mu_i^* y_i K(X_i, X_j))$$
- If we have a suitable Kernel function, we never need to compute $\phi(X)$.
- The range space of ϕ can even be infinite dimensional!



Example kernel function

- We start with an example kernel function in $\mathbb{R}^2$.
- Consider $K(X_i, X_j) = (1 + X_i^T X_j)^2$.
- Let $X_i = (x_{i1}, x_{i2})^T \in \mathbb{R}^2$ and similarly for X_j .
- Then

$$K(X_i, X_j) = (1 + x_{i1}x_{j1} + x_{i2}x_{j2})^2$$

- We now show that there exists a mapping ϕ such that $K(X_i, X_j) = \phi(X_i)^T \phi(X_j)$.



- Consider $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^6$ given by

$$Z = \phi(X) = [1 \quad \sqrt{2}x_1 \quad \sqrt{2}x_2 \quad x_1^2 \quad x_2^2 \quad \sqrt{2}x_1x_2]$$

(Here, $X = (x_1 \ x_2) \in \mathbb{R}^2$).

- It is easy to see that a linear discriminant function in terms of Z (i.e., in $\mathbb{R}^6$) would be a quadratic discriminant function in terms of X (i.e., in $\mathbb{R}^2$).
- Now we show that

$$K(X_i, X_j) = (1 + X_i^T X_j)^2 = Z_i^T Z_j = \phi(X_i)^T \phi(X_j)$$



- We have

$$\begin{aligned} Z_i^T Z_j &= 1 + 2x_{i1}x_{j1} + 2x_{i2}x_{j2} + x_{i1}^2x_{j1}^2 + x_{i2}^2x_{j2}^2 \\ &\quad + 2x_{i1}x_{i2}x_{j1}x_{j2} \\ &= (1 + x_{i1}x_{j1} + x_{i2}x_{j2})^2 \\ &= K(X_i, X_j) \end{aligned}$$

- Easy to see it works for $X \in \mathbb{R}^n$ in general.
- Thus $K(X_i, X_j) = (1 + X_i^T X_j)^2$ results in a quadratic discriminant function. 



- From this example, it is also easy to see that for a given Kernel function, the mapping ϕ (or the dimension of its range space) is not unique.
- Consider the same Kernel fn

$$K(X_i, X_j) = (1 + X_i^T X_j)^2.$$
- Consider the mapping $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}^7$ given by

$$Z = \phi(X) = [1 \quad \sqrt{2}x_1 \quad \sqrt{2}x_2 \quad x_1^2 \quad x_2^2 \quad x_1x_2 \quad x_1x_2]$$

- It is easy to see that this mapping also works.



Some Popular Kernel functions

- Polynomial kernel:

$$K_p(X_1, X_2) = (1 + X_1^T X_2)^p$$

- Gaussian kernel

$$K_G(X_1, X_2) = e^{-\frac{\|X_1 - X_2\|^2}{\sigma^2}}$$

- Sigmoidal kernel

$$K_S(X_1, X_2) = \tanh(aX_1^T X_2 + \theta)$$



SVM with Gaussian kernel

- Suppose we use $K_G(X_1, X_2) = e^{-\frac{\|X_1 - X_2\|^2}{\sigma^2}}$
- Classification of a new pattern X is determined by

$$\begin{aligned} f(X) &= \sum_{i \in S} \mu_i^* y_i K(X_i, X) + b^* \\ &= \sum_{i \in S} \mu_i^* y_i e^{-\frac{\|X_i - X\|^2}{2\sigma^2}} + b^* \end{aligned}$$

- Same as a Gaussian RBF (neural) network. Centers of hidden nodes are the support vectors.



SVM with Sigmoidal Kernel

- Classification of X is determined by sign of

$$\begin{aligned} f(X) &= \sum_{i \in S} \mu_i^* y_i K(X_i, X) + b^* \\ &= \sum_{i \in S} \mu_i^* y_i \tanh(a X^T X_i + \theta) + b^* \end{aligned}$$



- Same as the output of a three layer feedforward network with tanh activation function.
- There as many hidden nodes as support vectors and weights into the hidden node are given by the support vectors.

